INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, November-December, 2025

Degree: **B.Tech.** Semester: **First**

Stream: **CSBS/CSE/CSE(AI & ML)/CSE(AI)/ CSE(IOT)/ CSE(IOT, CSBT)/CSE(Cyber Security)/ CSIT/CST/CSE(Data Science)/CSE(Networks)/ ECE/EEE/EE/ME/IT/Robotics & AI**

Paper Code: **BSCBE104**

Paper Name: **Biology for Engineers**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. What is Brownian motion? **2**

2. What do you think, do we find Mendelian inheritance in Human beings too? **2**

3. Chromosomal mutation and gene more lethal gene mutation chromosome number increases mutation rate increase, is the statement true? **2**

4. Define competitive inhibition. **2**

5. Explain substrate saturation. **2**

6. When chemists in the laboratory want to make reactions occur faster they often use heat. To what extent could living organisms use this method to speed metabolic reactions? **2**

7. Describe the biochemical principles behind the use of the peptidases pepsin and papain as meat tenderizers. **2**

8. Why must the proteolytic enzymes produced in the pancreas and stomach be synthesized in an inactive form? **2**

9. Suggest a clinical treatment for lactose intolerance. **2**

10. Briefly describe two important functions of glycoproteins. **2**

11. Why are all monosaccharides and disaccharides soluble in water? **2**

12. Describe the forces that stabilize protein quaternary structure. **2**

13. When a protein is unfolded (denatured), it becomes less water soluble and often precipitates from solution. Explain. **2**

14. Explain pour plate technique. **2**

15. Define sterilization. **2**

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. What are the major aspects of biology? **5**

17. Draw similarities between eye and camera functions. **5**

18. Describe the mode of nutrition of the Five Kingdoms. **5**

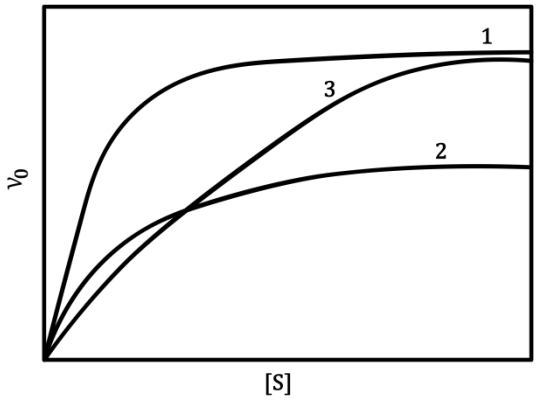
19. Explain the ultrastructure of prokaryotes and eukaryotes. **5**

20. Assume that an enzyme has the following kinetic constants: **5**

V_{max} = 50 µmol substrate transformed per minute Per milligram of enzyme
K_M = 0.0001 M

Calculate the substrate concentration that will yield a reaction rate of
(a) ½ V_{max}
(b) ⅓ V_{max}

21. Study the Michaelis-Menten graph below and identify the curve that corresponds to each reaction condition listed below. **5**
(a) No inhibitor
(b) Competitive inhibitor
(c) Noncompetitive inhibitor



22. Explain why haemoglobin is a better long-distance oxygen transporter than myoglobin. **5**

23. Study the following reaction, the hydrolysis of pyrophosphate: **5**

$$PP_i + H_2O \rightarrow 2\ P_i$$

(a) Write the reaction using structural formulas.
(b) What type of bond is cleaved?
(c) When is this reaction important in metabolism?

24. Write a short note on the ecological aspects of the single-celled organism. **5**

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Describe the differences among the actions of the three types of reversible inhibitors. Begin by identifying where each type of inhibitor binds to an enzyme. Your answer should include a comparison of the chemical structure of each type of inhibitor to the natural substrate and a discussion of the effect of each inhibitor on the V_{max} and K_m for the reaction. **10**

26. Derive the Lineweaver-Burk equation beginning with the Michaelis-Menten equation. **10**

27. In a laboratory experiment you completed a study of enzyme kinetics. The following data were collected: **10**

[S] (μmolar)	v_0 (μmolar/min)
50	10
100	19
150	31
200	38
300	55
400	62
800	68
1000	70

Estimate the K_M for this substrate: enzyme combination without graphing the data.

28. Compare amylose starch and glycogen in terms of the following characteristics. **10**

(a) Synthesized in what kind of organism?
(b) Biological role.
(c) Type of saccharide? mono, di, poly, etc.
(d) Chemical structure.
(e) Linkage between monosaccharide units.
(f) Type of branching.

29. Proteins may have up to four levels of structure. For each type, describe the kind of bonding interactions that maintain that type of structure. Select from covalent or noncovalent. If you choose noncovalent, also select specific types (hydrogen bonding, hydrophobic, ionic). **10**

(a) Primary
(b) Secondary
(c) Tertiary
(d) Quaternary

30. Compare the characteristics of globular and fibrous proteins. The following items should be discussed and contrasted: water solubility, amino acid composition, types of secondary and tertiary structure, biological function. **10**

31. Which of the following statements about mitochondrial respiration are true? Rewrite false statements to make them true. **10**

(a) The electron carriers involved in electron transport are present in the cytoplasm of the cell.
(b) NADH is a more powerful reducing agent than $FMNH_2$.
(c) The final electron acceptor is O_2.

(d) The electron carriers are arranged in order of increasing electron affinity.
(e) Fe^{2+} atoms in cytochromes and iron-sulfur proteins act as electron acceptors.
(f) $FADH_2$ can transfer one or two electrons to an acceptor.
(g) During electron transport, protons are pumped from the inner mitochondrial matrix to the other side of the membrane.

32. Briefly explain different phases of the bacterial growth curve. **10**

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